

Natural language statement

Domain

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Theorem 1.3.2. Let (X, μ) be a sigma-finite measure space, let (Y, ν) be another measure space, and let $0 < p_0 < p_1 \leq \infty$. Let T be a sublinear operator defined on $L^{p_0}(X) + L^{p_1}(X) = \{f_0 + f_1 : f_j \in L^{p_j}(X), j = 0, 1\}$ and taking values in the space of measurable functions on Y . Assume that there exist $A_0, A_1 < \infty$ such that $\|T(f)\|_{L^{p_0}(\nu)} \leq A_0 \|f\|_{L^{p_0}(X)}$ for all $f \in L^{p_0}(X)$, and $\|T(f)\|_{L^{p_1}(\nu)} \leq A_1 \|f\|_{L^{p_1}(X)}$ for all $f \in L^{p_1}(X)$. Then for all $p_0 < p < p_1$ and for all $f \in L^p(X)$ we have the estimate $\|T(f)\|_{L^p(Y)} \leq A \|f\|_{L^p(X)}$, where $A = 2[p/(p - p_0) + p/(p_1 - p)]^{1/p} A_0^{p/(p - p_0)} A_1^{p/(p_1 - p)}$.

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def IsSublinearOperator {X : Type u} {Y : Type v}
  (T : (X -> C) -> Y -> C) : Prop :=
  T 0 = 0 /\
  (forall f g x, ||T (f + g) x|| <= ||T f x|| + ||T g x||) /\
  forall c : C, forall f x, ||T (c * f) x|| = ||c|| * ||T f x||

def HasWeakType {X : Type u} {Y : Type v} [MeasurableSpace X]
[MeasurableSpace Y]
  (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> Y -> C)
  (p : R) (C : R>=0inf) : Prop :=
  forall f : X -> C, MemLp f (ENNReal.ofReal p) mu -> forall alpha : R, 0 <
  alpha ->
  nu {y | ENNReal.ofReal alpha < ||T f y||e} <=
  ENNReal.rpow (C * eLpNorm f (ENNReal.ofReal p) mu / ENNReal.ofReal
  alpha) p

def HasStrongType {X : Type u} {Y : Type v} [MeasurableSpace X]
[MeasurableSpace Y]
  (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> Y -> C)
  (p q : R) (C : R>=0inf) : Prop :=
  forall f : X -> C, MemLp f (ENNReal.ofReal p) mu ->
  MemLp (T f) (ENNReal.ofReal q) nu /\
  eLpNorm (T f) (ENNReal.ofReal q) nu <= C * eLpNorm f (ENNReal.ofReal
  p) mu

theorem grafakos_1_3_2_marcinkiewicz_interpolation
  {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y]
  (mu : Measure X) (nu : Measure Y) [SigmaFinite mu]
  (T : (X -> C) -> Y -> C) {p0 p1 p : R} {A0 A1 : R>=0inf}
  (hp : 0 < p0 /\ p0 < p /\ p < p1)
  (hA0 : A0 < inf) (hA1 : A1 < inf)
  (hT : IsSublinearOperator T) (h0 : HasWeakType mu nu T p0 A0)
  (h1 : HasWeakType mu nu T p1 A1) :
  HasStrongType mu nu T p p
  (2 * ENNReal.rpow
  (ENNReal.ofReal (p / (p - p0) + p / (p1 - p))) (1 / p) *
  ENNReal.rpow A0 ((p0 / p) * ((p1 - p) / (p1 - p0))) *
  ENNReal.rpow A1 ((p1 / p) * ((p - p0) / (p1 - p0)))) := by
  sorry
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Theorem 1.3.4. Let (X, μ) and (Y, ν) be two sigma-finite measure spaces. Let T be a linear operator defined on the set of all finitely simple functions on X and taking values in the set of measurable functions on Y . Let $1 \leq p_0, p_1, q_0, q_1 \leq \infty$ and assume that $\|T(f)\|_{L^{q_0}(Y)} \leq M_0 \|f\|_{L^{p_0}(X)}$ and $\|T(f)\|_{L^{q_1}(Y)} \leq M_1 \|f\|_{L^{p_1}(X)}$ for all finitely simple functions f on X . Then for all $0 < \theta < 1$ we have $\|T(f)\|_{L^q(Y)} \leq M_0^{1-\theta} M_1^\theta \|f\|_{L^p(X)}$ for all finitely simple functions f on X , where $1/p = (1-\theta)/p_0 + \theta/p_1$ and $1/q = (1-\theta)/q_0 + \theta/q_1$. Consequently, when $p < \infty$, by density, T has a unique bounded extension from $L^p(X, \mu)$ to $L^q(Y, \nu)$ when p and q are as above.

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def HasStrongType {X : Type u} {Y : Type v} [MeasurableSpace X]
[MeasurableSpace Y]
  (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> Y -> C)
  (p q : R) (C : R>=0inf) : Prop :=
  forall f : X -> C, MemLp f (ENNReal.ofReal p) mu ->
  MemLp (T f) (ENNReal.ofReal q) nu /\
  eLpNorm (T f) (ENNReal.ofReal q) nu <= C * eLpNorm f (ENNReal.ofReal
  p) mu

theorem grafakos_1_3_4_riesz_thorin_interpolation
  {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y]
  (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> l[C] (Y -> C))
  {p0 p1 q0 q1 p q theta : R} {M0 M1 : R>=0inf}
  (htheta : 0 < theta /\ theta < 1)
  (hexponents : 1 <= p0 /\ 1 <= p1 /\ 1 <= q0 /\ 1 <= q1)
  (hM0 : M0 < inf) (hM1 : M1 < inf)
  (hp : 1 / p = (1 - theta) / p0 + theta / p1)
  (hq : 1 / q = (1 - theta) / q0 + theta / q1)
  (h0 : HasStrongType mu nu T p0 q0 M0)
  (h1 : HasStrongType mu nu T p1 q1 M1) :
  HasStrongType mu nu T p q
  (ENNReal.rpow M0 (1 - theta) * ENNReal.rpow M1 theta) := by
  sorry
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Theorem 2.1.6. The uncentered and centered Hardy-Littlewood maximal operators M and M^c map $L^1(\mathbb{R}^n)$ to $L^{1,\infty}(\mathbb{R}^n)$ with constant at most 3^n and also $L^p(\mathbb{R}^n)$ to $L^p(\mathbb{R}^n)$ for $1 < p < \infty$ with constant at most $3^{n/p} p(p-1)^{-1}$. For any $f \in L^1(\mathbb{R}^n)$ we also have $\|M(f)\|_{L^p(\mathbb{R}^n)} \leq (3^n/\alpha) \int_{\mathbb{R}^n} |M(f)(y)| dy$.

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noncomputable def hardyLittlewoodMaximal (n : N)
  (f : EuclideanSpace R (Fin n) -> C) (X : EuclideanSpace R (Fin n)) :
  R>=0inf :=
  y : EuclideanSpace R (Fin n), r : {r : R // 0 < r},
  ( _ : x in ball y r),
  (integral^ z in ball y r, ||f z||e) / volume (ball y r)
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		<pre> noncomputable def hardyLittlewoodCenteredMaximal (n : N) (f : EuclideanSpace R (Fin n) -> C) (x : EuclideanSpace R (Fin n)) : R>=0inf := r : {r : R // 0 < r}, (integral^~ z in ball x r, f z e) / volume (ball x r) theorem grafakos_2_1_6_hardy_littlewood_maximal {n : N} : let operators := ({hardyLittlewoodMaximal n, hardyLittlewoodCenteredMaximal n} : Set ((EuclideanSpace R (Fin n) -> C) -> EuclideanSpace R (Fin n) -> R>=0inf)) (forall M in operators, forall f : EuclideanSpace R (Fin n) -> C, MemLp f 1 volume -> forall alpha : R, 0 < alpha -> volume {x ENNReal.ofReal alpha < M f x} <= ENNReal.ofReal (3 ^ n / alpha) * integral^~ x in {x ENNReal.ofReal alpha < M f x}, f x e) /\ forall M in operators, forall p : R, 1 < p -> forall f : EuclideanSpace R (Fin n) -> C, MemLp f (ENNReal.ofReal p) volume -> ENNReal.rpow (integral^~ x, ENNReal.rpow (M f x) p) (1 / p) <= ENNReal.ofReal (3 ^ ((n : R) / p) * p / (p - 1)) * eLpNorm f (ENNReal.ofReal p) volume := by sorry </pre>
Theorem 2.2.14. Given f, g, and h in $S(\mathbb{R}^n)$, we have (1) $\int \mathbb{R}^n f(x)g(x) \, dx = \int \mathbb{R}^n f(x)g(x) \, dx$; (2) (Fourier Inversion) $(fV) = f = (fV)$; (3) (Parseval's relation) $\int \mathbb{R}^n f(x)h(x) \, dx = \int \mathbb{R}^n f(x)h(x) \, dx$; (4) (Plancherel's identity) $\ f\ _{L^2} = \ f\ _{L^2}$; (5) $\int \mathbb{R}^n f(x)h(x) \, dx = \int \mathbb{R}^n f(x)hV(x) \, dx$.	Fourier analysis	<pre> theorem grafakos_2_2_14_fourier_identities_on_schwartz {n : N} (f g h : S(EuclideanSpace R (Fin n), C)) : ((integral x, f x * F g x) = integral x, F f x * g x) /\ F^~ (F f) = f /\ F (F^~ f) = f /\ (integral x, star (F f x) * F g x) = integral x, star (f x) * g x /\ eLpNorm (fun x => F f x) 2 volume = eLpNorm f 2 volume /\ eLpNorm (fun x => F^~ f x) 2 volume = eLpNorm f 2 volume /\ (integral x, F f x * h x) = integral x, f x * F h x := by sorry </pre>
Proposition 2.2.16 (Hausdorff-Young inequality). For every function f in $L_p(\mathbb{R}^n)$ we have the estimate $\ f\ _{p'} \leq \ f\ _p$ whenever $1 \leq p \leq 2$.	Fourier analysis	<pre> theorem grafakos_2_2_16_hausdorff_young {n : N} {p : R} {f : EuclideanSpace R (Fin n) -> C} (hp : 1 <= p /\ p <= 2) (hf : MemLp f (ENNReal.ofReal p) volume) : let conjugateExponent : R>=0inf := if p = 1 then inf else ENNReal.ofReal (p / (p - 1)) MemLp (F f) conjugateExponent volume /\ eLpNorm (F f) conjugateExponent volume <= eLpNorm f (ENNReal.ofReal p) volume := by sorry </pre>
Theorem 3.2.8 (Poisson summation formula). Let f be a continuous function on $\mathbb{R}^n$ which satisfies $f(x) \leq C(1+ x)^{-n-\delta}$ for some $C, \delta > 0$ and whose Fourier transform $\hat{f}$ restricted on $\mathbb{Z}^n$ satisfies $\sum_{m \in \mathbb{Z}^n} \hat{f}(m) < \infty$. Then for all x in $\mathbb{R}^n$, $\sum_{m \in \mathbb{Z}^n} f(m)e^{2\pi i m \cdot x} = \sum_{k \in \mathbb{Z}^n} f(x+k)$, and in particular $\sum_{m \in \mathbb{Z}^n} f(m) = \sum_{k \in \mathbb{Z}^n} f(k)$.	Fourier analysis	<pre> theorem grafakos_3_2_8_poisson_summation {n : N} {f : EuclideanSpace R (Fin n) -> C} (hf : Continuous f /\ Integrable f /\ (exists C delta : R, 0 < C /\ 0 < delta /\ forall x, f x <= C * (1 + x) ^ (-(n : R) - delta)) /\ Summable fun m : Fin n -> Z => F f ((WithLp.toLp 2 fun i => (m i : R))))) : (forall x : EuclideanSpace R (Fin n), sum' m : Fin n -> Z, F f ((WithLp.toLp 2 fun i => (m i : R)))) * Complex.exp (2 * Real.pi * Complex.I * (sum i, (m i : C) * (x i : C))) = sum' k : Fin n -> Z, f (x + (WithLp.toLp 2 fun i => (k i : R)))) /\ (sum' m : Fin n -> Z, F f ((WithLp.toLp 2 fun i => (m i : R)))) = sum' k : Fin n -> Z, f ((WithLp.toLp 2 fun i => (k i : R))) := by sorry </pre>
Theorem 4.1.1. For $R > 0$ and m in $\mathbb{Z}^n$, let $a(m, R)$ be complex numbers such that (i) for every $R > 0$ there is a qR such that $a(m, R) = 0$ if $m > qR$; (ii) there is an $M_0 < \infty$ such that $a(m, R) \leq M_0$ for all m in $\mathbb{Z}^n$ and all $R > 0$; (iii) for each m in $\mathbb{Z}^n$, the limit of $a(m, R)$ exists as $R \rightarrow \infty$ and $\lim_{R \rightarrow \infty} a(m, R) = a_m$. Let $1 \leq p < \infty$. For f in $L_p(\mathbb{T}^n)$ and x in $\mathbb{T}^n$ define $S_R(f)(x) = \sum_{m \in \mathbb{Z}^n} a(m, R)f(m)e^{2\pi i m \cdot x}$, noting that the sum is well defined because of (i). Also, for h in $C_{inf}(\mathbb{T}^n)$ define $A(h)(x) = \sum_{m \in \mathbb{Z}^n} a_m h(m)e^{2\pi i m \cdot x}$. Then	Fourier analysis	<pre> def HasStrongType {X : Type u} {Y : Type v} [MeasurableSpace X] [MeasurableSpace Y] (mu : Measure X) (nu : Measure Y) (T : (X -> C) -> Y -> C) (p q : R) (C : R>=0inf) : Prop := forall f : X -> C, MemLp f (ENNReal.ofReal p) mu -> MemLp (T f) (ENNReal.ofReal q) nu /\ eLpNorm (T f) (ENNReal.ofReal q) nu <= C * eLpNorm f (ENNReal.ofReal p) mu </pre>

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for all f in $L_p(T_n)$ the sequence $S_n R(f)$ converges in L_p as $R \rightarrow \infty$ if and only if there exists a constant $K < \infty$ such that $\sup_{R>0} \ S_n R\ _{L_p \rightarrow L_p} \leq K$. Furthermore, if this holds, then for the same constant K we have $\sup_{\{h \in C_0^\infty, h \neq 0\}} \ A(h)\ _{L_p} / \ h\ _{L_p} \leq K$, and then A extends to a bounded operator A from $L_p(T_n)$ to itself; moreover, for every f in $L_p(T_n)$ we have that $S_n R(f) \rightarrow A(f)$ in L_p as $R \rightarrow \infty$.		<pre> noncomputable def torusCharacter {n : ℕ} (m : Fin n → ℤ) (x : Fin n → AddCircle (1 : ℝ)) : ℝ := prod i, fourier (m i) (x i) noncomputable def torusFourierCoefficient {n : ℕ} [MeasurableSpace (Fin n → AddCircle (1 : ℝ))] (mu : Measure (Fin n → AddCircle (1 : ℝ))) (f : (Fin n → AddCircle (1 : ℝ)) → ℝ) (m : Fin n → ℤ) : ℝ := integral x, star (torusCharacter m x) * f x dmu theorem grafakos_4_1_1_torus_summability_uniform_boundedness {n : ℕ} (a : ℝ → (Fin n → ℤ) → ℝ) (aLimit : (Fin n → ℤ) → ℝ) (hfinite : forall R, 0 < R → (Function.support (a R)).Finite) (hbounded : exists M : ℝ, 0 ≤ M ∧ forall R m, 0 < R → a R m ≤ M) (htendsto : forall m, Tendsto (fun R => a R m) atTop (nhds (aLimit m))) {p : ℝ} (hp : 1 ≤ p) : let mu : Measure (Fin n → AddCircle (1 : ℝ)) := volume let S := fun R (f : (Fin n → AddCircle (1 : ℝ)) → ℝ) (x : Fin n → AddCircle (1 : ℝ)) => sum' m, a R m * torusFourierCoefficient mu f m * torusCharacter m x let A := fun (f : (Fin n → AddCircle (1 : ℝ)) → ℝ) (x : Fin n → AddCircle (1 : ℝ)) => sum' m, aLimit m * torusFourierCoefficient mu f m * torusCharacter m x ((forall f, MemLp f (ENNReal.ofReal p) mu → Tendsto (fun R => eLpNorm (S R f - A f) (ENNReal.ofReal p) mu) atTop (nhds 0)) <=> exists C : ℝ => 0, C < ∞ ∧ forall R, 0 < R → HasStrongType mu mu (S R) p p C ∧ forall C : ℝ => 0, (forall R, 0 < R → HasStrongType mu mu (S R) p p C) → HasStrongType mu mu A p p C := by sorry </pre>
Theorem 4.3.15. For every $1 < p < \infty$ there exists a finite constant C_p such that for all f in $C_0^\infty(\mathbb{R})$ we have $\ C^{**}(f)\ _{L_p(\mathbb{R})} \leq C_p \ f\ _{L_p(\mathbb{R})}$, where $C^{**}(f)(x) = \sup_{R>0} \int_{ x \leq R} f(x) e^{2\pi i x} dx$ is the Carleson operator.	Fourier analysis	<pre> noncomputable def carlesonHuntMaximal (f : ℝ → ℝ) (x : ℝ) : ℝ := R : {R : ℝ // 0 < R}, integral xi in Set.Icc (-R.1) R.1, F f xi * Complex.exp (2 * Real.pi * Complex.I * xi * x) e theorem grafakos_4_3_15_carleson_hunt_line {p : ℝ} (hp : 1 < p) : exists C : ℝ => 0, C < ∞ ∧ forall f : ℝ → ℝ, ENNReal.rpow (integral^~ x, ENNReal.rpow (carlesonHuntMaximal f x) p) (1 / p) ≤ C * eLpNorm (f : ℝ → ℝ) (ENNReal.ofReal p) volume := by sorry </pre>
Theorem 5.3.1. Let f in $L_1(\mathbb{R}^n)$ and $\alpha > 0$. Then there exist functions g and b on $\mathbb{R}^n$ such that $f = g + b$; $\ g\ _{L_1} \leq \ f\ _{L_1}$ and $\ g\ _{L_1} \leq 2^{n\alpha}$; $b = \sum_j b_j$ where each b_j is supported in a dyadic cube Q_j, the cubes are disjoint, $\int_{Q_j} b_j(x) dx = 0$, $\ b_j\ _{L_1} \leq 2^{(n+1)\alpha} Q_j$, and $\sum_j Q_j \leq \alpha^{-(n+1)} \ f\ _{L_1}$.	Fourier analysis	<pre> structure DyadicCube (n : ℕ) where scale : ℤ corner : Fin n → ℤ def DyadicCube.carrier {n : ℕ} (Q : DyadicCube n) : Set (EuclideanSpace ℝ (Fin n)) := {x forall i, (Q.corner i : ℝ) * 2 ^ Q.scale ≤ x i ∧ x i < (Q.corner i + 1 : ℤ) * 2 ^ Q.scale} theorem grafakos_5_3_1_calderon_zygmund_decomposition {n : ℕ} {f : EuclideanSpace ℝ (Fin n) → ℝ} {alpha : ℝ} (hf : MemLp f 1 volume) (halpha : 0 < alpha) : exists g b : EuclideanSpace ℝ (Fin n) → ℝ, exists (Q : ℕ → DyadicCube n) (bad : ℕ → EuclideanSpace ℝ (Fin n) → ℝ), f = g + b ∧ HasSum bad b ∧ eLpNorm g 1 volume ≤ eLpNorm f 1 volume ∧ eLpNorm g inf volume ≤ ENNReal.ofReal (2 ^ n * alpha) ∧ (Pairwise fun i j => Disjoint (Q i).carrier (Q j).carrier) ∧ (forall j, Function.support (bad j) subset (Q j).carrier) ∧ (forall j, integral x, bad j x = 0) ∧ (forall j, eLpNorm (bad j) 1 volume ≤ ENNReal.ofReal (2 ^ (n + 1) * alpha) * volume (Q j).carrier) ∧ </pre>

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		<pre>sum' j, volume (Q j).carrier <= eLpNorm f 1 volume / ENNReal.ofReal alpha := by sorry</pre>
Theorem 5.6.6. For $1 < p, r < \infty$ the Hardy-Littlewood maximal function M satisfies the vector-valued inequalities $\ (\sum_j M(f_j) ^r)^{1/r} \ _{1, \infty} \leq C_n (1 + (r-1)^{-1}) \ (\sum_j f_j ^r)^{1/r}\ _{1, \infty}$ and $\ (\sum_j M(f_j) ^p)^{1/p} \ _{p, r} \leq C_n c(p, r) \ (\sum_j f_j ^p)^{1/p}\ _{p, r}$.	Fourier analysis	<pre>noncomputable def hardyLittlewoodMaximal (n : ℕ) (f : EuclideanSpace ℝ (Fin n) → ℝ) (x : EuclideanSpace ℝ (Fin n)) : ℝ := y : EuclideanSpace ℝ (Fin n), r : {r : ℝ // 0 < r}, (_ : x in ball y r), (integral^~ z in ball y r, f z ^e) / volume (ball y r) theorem grafakos_5_6_6_vector_valued_maximal {n : ℕ} {p r : ℝ} (hp : 1 < p) (hr : 1 < r) : let ellNorm := fun (f : ℕ → EuclideanSpace ℝ (Fin n) → ℝ) x => ENNReal.rpow (sum' j, ENNReal.rpow f j x ^e r) (1 / r) let maximalNorm := fun (f : ℕ → EuclideanSpace ℝ (Fin n) → ℝ) x => ENNReal.rpow (sum' j, ENNReal.rpow (hardyLittlewoodMaximal n (f j) x) r) (1 / r) exists Cn Cp : ℝ := 0, Cn < inf /\ Cp < inf /\ (forall f : ℕ → EuclideanSpace ℝ (Fin n) → ℝ, forall alpha : ℝ, 0 < alpha -> volume {x ENNReal.ofReal alpha < maximalNorm f x} <= Cn * ENNReal.ofReal (1 + 1 / (r - 1)) / ENNReal.ofReal alpha * ENNReal.rpow (integral^~ x, ellNorm f x) 1) /\ forall f : ℕ → EuclideanSpace ℝ (Fin n) → ℝ, ENNReal.rpow (integral^~ x, ENNReal.rpow (maximalNorm f x) p) (1 / p) <= Cp * ENNReal.rpow (integral^~ x, ENNReal.rpow (ellNorm f x) p) (1 / p) := by sorry</pre>